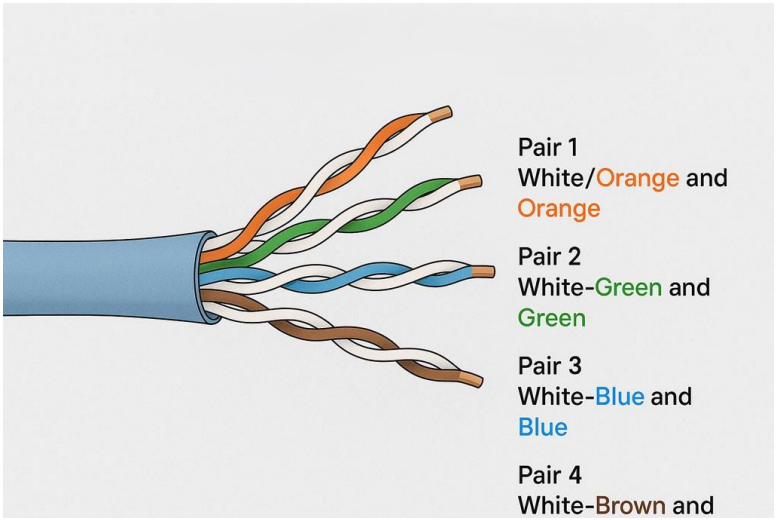
Network Cable and connectors:-

Tasks :-

1.Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Ans. :-



RJ-45 Pin Wire Color :-

1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Key Points:-

- Cable Type: Cat6 Ethernet Cable
- Connector: RJ-45 (8P8C)
- Standard Used: T568B
- Number of Wires: 8 conductors arranged as 4 twisted pairs
- Purpose: Used for Gigabit Ethernet (1 Gbps) and up to 10 Gbps over shorter distances (up to 55 m for 10GBASE-T).

2.List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Ans.:-

Cat5e Ethernet Cable :-

- **WiFi Router (Home/Office):-**
  It Is Used :-

  Cat5e cables support speeds up to 1 Gbps over 100 meters, making them suitable for connecting a WiFi router to a modem or desktop computer. They are affordable and reliable for everyday internet use.

Cat6 Ethernet Cable :-

- **Zomato POS Terminal (Restaurant Billing System) :-**
  It Is Used :-
  Cat6 cables support 1–10 Gbps speeds with better protection against interference than Cat5e. Restaurants use them to connect POS terminals, billing systems, and payment devices for fast and stable network communication.

Fiber Optic Cable :-

- **IPL Streaming Server (Data Center) :-**
  It Is Used :-
  Fiber optic cables transmit data using light, providing very high speeds (10 Gbps and above) over long distances with minimal signal loss. They are ideal for streaming live IPL matches to millions of viewers because they offer high bandwidth and reliability.

3.Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Ans.:-

Step-by-Step Process for Crimping an RJ-45 Connector onto a Network Cable :-

1. Cut the Ethernet cable to the required length using a cable cutter.
2. Strip the outer jacket (about 2–3 cm) using a wire stripper, taking care not to damage the inner wires.
3. Untwist the four wire pairs and straighten all eight wires with your fingers.
4. Arrange the wires in the correct order according to the T568B standard:
   - White/Orange
   - Orange
   - White/Green
   - Blue
   - White/Blue
   - Green
   - White/Brown
   - Brown
5. Trim the wires evenly so that all eight wires are the same length (about 12–13 mm from the cable jacket).
6. Insert the wires into the RJ-45 connector, ensuring each wire reaches the end of its slot and remains in the correct order. The cable jacket should also extend slightly inside the connector.

7. Place the connector into the RJ-45 crimping tool and squeeze the handles firmly until the connector is fully crimped.
8. Test the cable using a network cable tester to confirm all eight pins are connected correctly and there are no wiring faults.

Two Common Mistakes to Avoid :-

1. Incorrect wire order
    - Mixing up the wire sequence (such as not following the T568B standard) can prevent the cable from working properly or reduce network performance.
2. Not pushing the cable jacket into the connector
    - If only the individual wires are crimped and the outer jacket is left outside the connector, the cable becomes weak and may disconnect or break with regular use.

A properly crimped RJ-45 connector provides a secure, reliable Ethernet connection for devices such as computers, routers, switches, and network printers.

4.Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Ans.:-

Scenario: Setting Up a Gaming Zone for an IPL Screening Event :-

To provide a fast, stable, and reliable network for multiple gaming consoles and large display screens, I would use Cat6 Ethernet cables with RJ-45 connectors for all device connections. If the venue is very large and the networking equipment is far apart, I would use fiber optic cables for the main backbone connection between network switches.

Connection	Cable Type	Connector	Reason
Router to Network Switch	Cat6 Ethernet	RJ-45	Supports Gigabit and up to 10 Gbps speeds, providing a fast and stable connection.
Switch to Gaming Consoles	Cat6 Ethernet	RJ-45	Reduces lag and packet loss, ensuring smooth multiplayer gaming.
Switch to Smart TVs/Streaming Screens	Cat6 Ethernet	RJ-45	Provides reliable, uninterrupted streaming of the IPL match in high definition.
Main Backbone (Long Distance)	Fiber Optic Cable	LC or SC Connector	Offers very high bandwidth, long-distance transmission, and immunity to electrical interference.

These Choices :-

- **Cat6 Ethernet Cable :-**
  - Supports speeds up to 10 Gbps over shorter distances.
  - Provides low latency, which is essential for online gaming.
  - More resistant to interference than Cat5e, ensuring stable performance.
- **RJ-45 Connector ::-**
  - Standard connector for Ethernet cables.
  - Easy to install and compatible with routers, switches, gaming consoles, PCs, and smart TVs.
- **Fiber Optic Cable (for Backbone):-**
  - Ideal for connecting network equipment over long distances.
  - Delivers extremely high speed with minimal signal loss.
  - Suitable for handling heavy network traffic during an IPL screening event.

**5.Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.**

Ans.:-

Cat6 Ethernet Cable with RJ-45 Connector :-



Cable Type: Cat6 Ethernet cable

Connector Type: RJ-45 (8P8C) connector

Devices Connected:

- WiFi Router - Network Switch
- WiFi Router - Desktop Computer
- Network Switch - Gaming Console or Smart TV

The images show a Cat6 Ethernet cable with a transparent RJ-45 connector plugged into a router or network switch. The RJ-45 connector has eight metal contacts that carry data over the Ethernet cable. Cat6 cables are commonly blue, yellow, or gray and are designed for high-speed wired networking.

Used:-

- Supports speeds up to 1 Gbps over 100 m and up to 10 Gbps over shorter distances.
- Provides a stable, low-latency connection for internet access.
- Commonly used in homes, colleges, offices, cafés, and gaming zones to connect routers, switches, PCs, POS terminals, and smart TVs.